SnapAxis: Axis Based Interaction for Object Manipulation in Mixed Reality

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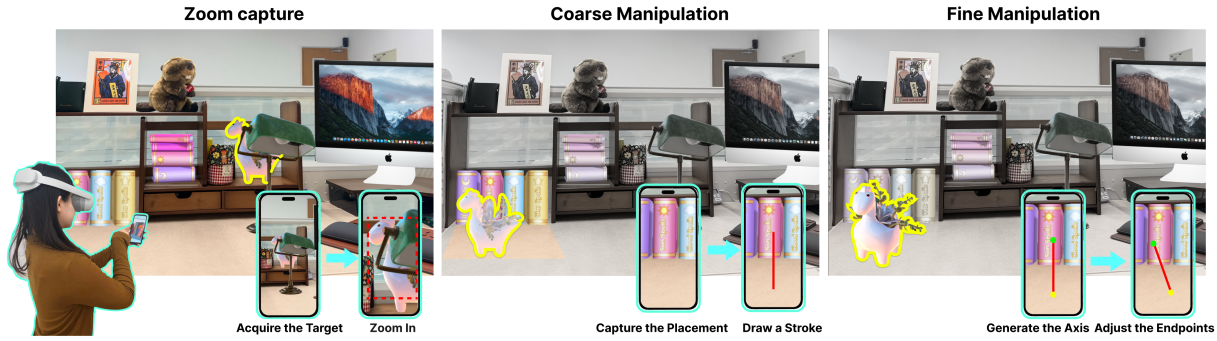


Figure 1: Overview of SnapAxis. Users first acquire the target with ZoomCapture, then establish placement and axis direction in Coarse Manipulation, and finally refine the object through endpoint based Fine Manipulation.

Abstract

Accurate 3D object manipulation in mixed reality remains difficult when targets are distant or partially occluded. Existing techniques often improve local stages such as target acquisition or fine adjustment, but pay less attention to maintaining a stable and interpretable manipulation reference across acquisition, placement, and refinement. This forces users to repeatedly reinterpret control mappings, increasing cognitive load and reducing refinement stability. We present SnapAxis, an Axis based Interaction framework for MR object manipulation through smartphone. SnapAxis is organized around the Semantic Axis Handle, a shared object centered reference that links acquisition, placement, and refinement. The framework consists of three stages: ZoomCapture for target acquisition, Coarse Manipulation for Semantic Axis Handle establishment, and Fine Manipulation for continued refinement around the established reference. A within subject user study showed that SnapAxis significantly improved task efficiency and 3D alignment accuracy while reducing phone movement, head view search burden, and subjective workload.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**.

Keywords

Mixed reality, MR input, interaction techniques, 3D controller

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1 Introduction

Mixed reality (MR) enables users to place and manipulate virtual objects within real world environments. However, accurate 3D object manipulation remains challenging when targets are distant or partially occluded. In such scenarios, objects often occupy only a small portion of the user's view, making selection ambiguous and depth relationships difficult to interpret. These issues are further amplified when users need to perform precise alignment involving translation, rotation, and scaling.

Prior work has improved MR object interaction through ray based reach extension, smartphone based input [26], and occlusion aware target disambiguation[52]. However, they often treat target acquisition, coarse placement, and fine manipulation as loosely connected stages, each relying on different control strategies or spatial references. As a result, users must repeatedly reinterpret how input maps to object position, orientation, and scale across stages, rather than manipulating around a stable shared reference.

This issue is especially important for smartphone based MR interaction. Smartphones are highly familiar devices, and users already know how to perform operations such as pinch to zoom and tapping to select. In MR, smartphones are visible to the user, presenting a novel approach for MR interaction. Prior work has shown that smartphones can support 3D object manipulation through combinations of motion sensing[18], touch input[42], camera based view mediation or environment aware adaptation[46]. However, these approaches often treat the smartphone as a generic six degree of freedom (DOF) surrogate.

Compared with a conventional controller, a smartphone provides a current view that can be captured and reused as a manipulation reference, together with familiar touch input and pose aware interaction capabilities, making it a suitable medium for establishing and controlling object centered references. In our work, the value of the smartphone lies not merely in serving as a generic 6DoF surrogate, but in supporting a view grounded interaction reference that can link target acquisition and subsequent manipulation.

Despite this potential, three challenges remain in using smartphones for 3D object manipulation in distant and highly occluded scenes. 1) The small physical size of the smartphone screen limits the exposed area of distant or occluded objects, making accurate selection difficult. 2) It is difficult to simultaneously support the efficiency of large scale movement and the accuracy of small scale adjustment, and existing interactions often struggle to transition smoothly between the two. 3) Depth related manipulation still lacks sufficiently intuitive and interpretable references, especially under single view and occluded conditions. Overall, these challenges suggest that the difficulty is not only target visibility or control precision in isolation, but how to establish a stable object centered reference early and preserve it across stages.

To address these challenges, SnapAxis organizes interaction around a shared object centered reference, the Semantic Axis Handle (SAH), which remains meaningful across acquisition, placement, and refinement. To operationalize this idea, SnapAxis consists of three connected stages. It first uses ZoomCapture to support target search, localization, and confirmation by enlarging visible target evidence before commitment. It then uses Coarse Manipulation to establish and externalize the SAH through placement, axis sketching, and depth adjustment. Finally, it supports Fine Manipulation, in which users continue refining translation, rotation, and scaling around the same established reference rather than switching to unrelated control mappings. We evaluate SnapAxis through a controlled user study against a conventional ray based technique and a smartphone based baseline. Results show that SnapAxis significantly improves efficiency and accuracy while reducing phone movement, horizontal view search burden, and subjective workload. Figure 1 illustrates the process of using SnapAxis to move a dinosaur from the bookshelf to the desk.

In summary, the contributions of our method are as follows.

- We present SnapAxis, an axis based interaction framework for MR object manipulation under distance and occlusion.
- We introduce the Semantic Axis Handle, an object centered and directionally meaningful manipulation reference that can be established early and preserved across target acquisition, coarse placement, and fine refinement.
- We evaluate SnapAxis in a controlled user study, showing significant improvements in alignment accuracy, efficiency, and workload over a ray based technique and a smartphone based baseline.

2 Related Work

In this section, we review prior research briefly on distant and occluded object manipulation in MR, smartphone based controllers, and axis oriented and DOF separated manipulation. For a broader overview, we also refer readers to existing survey papers on object manipulation in VR [2, 32].

2.1 Distant and Occluded Object Manipulation in MR

Prior work on near and far field interaction has established a rich design space for object manipulation in immersive environments. In near field settings, where targets are within arm's reach, direct techniques such as grab and move[59], virtual handles[26]provide strong spatial correspondence and immediate visual feedback. These approaches benefit from proprioceptive cues and continuous spatial coupling between user action and object motion, allowing users to perform translation and rotation with relatively low cognitive effort.

As interaction distance increases, direct manipulation becomes impractical. To support far field interaction, prior work has proposed ray based pointing[4], proxy objects[40],to enlarge the reachable space, and later work[55] adapts control display gain so that the same input serves both gross transport and fine adjustment. World in miniature representations[45]shrinks the entire scene into a handheld replica, but the scale compression amplifies small input errors once they are mapped back to full scale. Later designs bring only the target within reach and mitigate this amplification, yet each introduces its own cost. SightWarp[30] requires an explicit summoning step and repeated refocusing between near and far views. These methods effectively extend the user's reach beyond physical constraints, but often introduce depth ambiguity, reduced

control stability, and stronger reliance on visual estimation. In particular, when targets are distant or partially occluded, users often need to act on limited visible evidence, making it difficult to confidently acquire the target before manipulation begins. As a result, the challenge is not only how to reach the object, but also how to establish a sufficiently reliable reference for subsequent placement and refinement.

To bridge near and far field interaction, prior work has proposed techniques that dynamically adjust control display gain, rescale motion, or introduce explicit transitions between interaction modes[12, 41, 54, 59]. Other approaches switch reference frames or viewpoints by warping space or introducing proxies, effectively re anchoring manipulation around the user's current view[1, 30]. But, viewpoint cloning[29] can introduce visuomotor remapping because the displayed representation is spatially displaced from the actual manipulation site. VMirror[28] adds placement effort while smaller, distorted reflections increase selection errors. Other works structures interaction into multiple stages, typically combining coarse positioning with subsequent fine refinement[48]. These techniques improve accessibility and reduce gross error for distant targets, but they often rely on mode switches, nonlinear mappings, or auxiliary representations that require users to reinterpret how their input maps to object motion as task conditions change.

Overall, these observations suggest that the challenge of distant and occluded manipulation is not only how to extend reach, but also how to maintain a sufficiently stable reference for subsequent placement and refinement as spatial conditions evolve.

2.2 Smartphone based controllers

A growing body of work explores smartphones as input devices for XR, treating them as tracked controllers that combine high resolution touch input with pose sensing. Prior works fuse device pose with touch input to enable bimanual manipulation in constrained settings [26], combine device movement with peripheral touch to extend the reachable range of device based control on handheld MR displays[31] and can complement head mounted displays by providing precise selection, parameter adjustment, and interface offloading through cross device interaction [58]. More recent frameworks further push consistency across access points (handheld MR, MR HMD, VR HMD) by reusing tablet based control primitives across devices [31, 47]. Together, these studies demonstrate that smartphones are not merely auxiliary displays, but versatile interaction media that can support both input and spatial control in immersive environments. Beyond serving as generic controllers, smartphones also function as tangible handheld artifacts with familiar physical affordances. Prior work on mobile haptic interfaces and handheld tool metaphors suggests that everyday devices can support embodied interaction by allowing users to appropriate grip, touch, and motion as part of spatial tasks[53, 57].

Precision miniature control paradigms further show that handheld tools can facilitate fine manipulation in immersive environments by leveraging stable physical support and high input fidelity[18, 36, 38]. These findings make smartphones especially relevant for tasks that require both rapid interaction and precise adjustment.

Despite the significant advantages of smartphones as controllers in MR environments, they still face several limitations, the most prominent being the small screen size, which restricts the visibility of distant or occluded objects, making precise object interaction difficult[10, 16, 17]. As a result, although smartphones provide a powerful combination of view, touch, and pose, they are often still used as generic 6DoF surrogates rather than as devices that can actively structure the relationship between target acquisition and subsequent manipulation. In our view, this reveals an important design opportunity: instead of treating the smartphone only as an input substitute, it can be used as a medium for linking target acquisition and later object manipulation under a shared object centered reference. This motivates our use of the smartphone not simply as a controller, but as a device through which a stable manipulation reference can be established and reused across stages.

2.3 Axis Oriented and DoF Separated Manipulation

Numerous studies have shown that reducing or structuring degrees of freedom (DoFs) can significantly enhance the performance of 3D object manipulation [22, 25]. Axial based manipulation, by constraining movement to a single interpretable direction, has become a widely adopted and effective technique for achieving efficient control and stability. Limiting motion to a single axis allows users to perform operations such as rotation, translation, and scaling with greater precision [27, 33, 35, 50]. Alternative approaches use custom transformation axes or axis aligned control handles to stabilize operations and improve precision [34, 44]. Together, these methods show that object manipulation often benefits from references that are more structured and interpretable than unconstrained full DoF control.

However, these approaches still face limitations when 2D input must be mapped onto 3D operations. In particular, when users need to adjust object depth or perform multi DoF manipulation, traditional single axis control may lack flexibility, leading to unintuitive interaction and increased error. Likewise, although DoF separation can reduce interference among control dimensions, it often requires users to switch repeatedly among multiple axes or control states, increasing cognitive load and operational complexity. These limitations become especially pronounced in distant or occluded scenes, where users already face restricted visibility and must continuously reestablish their operating reference during interaction. From this perspective, the value of axis oriented and DoF separated manipulation lies not only in improving precision, but also in providing structured and interpretable references for interaction. This suggests a broader design opportunity: rather than treating axis as isolated control constraints, interaction can be organized around a persistent object centered handle that remains meaningful from early placement to later refinement.

2.4 Summary

Across these threads, prior work has examined near far field control, smartphone based input, and axis or DoF structured manipulation largely in isolation. Techniques that improve distant access through input modality transitions or proxy summoning still treat selection, coarse placement, and fine refinement as separate episodes. None of them maintains an object centered reference that is established at acquisition and remains stable through refinement, so users must

repeatedly reacquire the target or fall back on a frame that is not anchored to the object. The question we address is therefore how an everyday device can support cross device manipulation of distant and occluded virtual objects in MR without breaking this reference.

3 Method

In this section, we first present the design guidelines for smartphone based interaction in MR, focusing on the manipulation of virtual objects in 3D space, particularly in distant and occluded scenarios (3.1). We then introduce the Semantic Axis Handle (3.2) and Framework (3.3) derived from these guidelines. The following sections describe the major states of the framework, including ZoomCapture (3.4), coarse manipulation (3.5), and fine manipulation (3.6).

3.1 Design Guidelines

Smartphone based manipulation of virtual objects in MR faces three recurring challenges: limited visible target exposure, restricted flexibility for 3D interaction on small screens, and the difficulty of balancing large-scale movement with small scale precision. To address these issues, we derived a set of design guidelines for smartphone based object manipulation. These guidelines were informed by prior work on smartphone, far field, and axis based interaction, as well as formative observations and iterative prototyping during the development of SnapAxis.

1. Touchscreen Input as an Intuitive Spatial Proxy: Smartphones provide an interaction surface that users already understand as spatial and intentional. Prior work has shown that smartphones can serve as effective spatial input devices for XR by combining high resolution touch input with device pose and motion sensing, rather than functioning merely as generic handheld controllers[8, 37, 49] In our early pilot tasks, users naturally interpreted smartphone actions such as framing, snapshot capture, drawing, dragging, and tapping as object oriented operations for locating, binding, selecting, and repositioning virtual content. These mappings are consistent with mental models[9, 42] developed through everyday mobile interaction. This suggests that the smartphone can serve not only as an input device, but also as a shared manipulation reference that links target acquisition to subsequent object manipulation.

2. Object Inherent Directionality as a Natural Manipulation Cue: Our observations suggest that users do not reason about all manipulation dimensions in the same way. Translation is often interpreted relative to global space, whereas rotation and scaling are more often understood relative to the object itself. This indicates that manipulation handles should accommodate both global and local frames of reference[14, 24]. In addition, users often infer an initial directional intention from the object's own structure. Objects with elongated shapes, salient principal faces, or functional asymmetry often implicitly suggest a principal direction. Even for geometrically symmetric objects such as cubes, users still tend to construct an object local coordinate frame that is related to the current viewpoint. Within this local frame, one of the directions parallel to the current view is often treated as the principal direction. Designing manipulation handles around this principal direction therefore offers strong intuitiveness and interpretability. From our formative observations, we found that in early manipulation stages, specifying an axis through a sketched stroke was the most natural and behaviorally consistent way for users to externalize directional intent and establish a reference for subsequent interaction. This suggests that effective 3D manipulation should combine global spatial reference for overall layout with object centered directional structure for refinement, rather than relying solely on abstract world axis[15].

3. Coarse to Fine Interaction with Stable References: MR manipulation often spans multiple spatial scales. Users may first need to make large adjustments at a distance and then perform precise alignment at close range. Prior work has shown that users naturally adopt such coarse to fine strategies in 3D interaction[4, 5]. Our observations were consistent with this pattern. Coarse manipulation was more effective for rapidly reducing large spatial errors, whereas fine manipulation was critical for avoiding overshoot and instability. This suggests that MR interaction should support gradual refinement while preserving consistency across interaction stages.

3.2 Semantic Axis Handle

We introduce the Semantic Axis Handle (SAH) as a persistent object centered reference for manipulation. Rather than treating manipulation as a sequence of unrelated control mappings, the SAH provides a stable directional handle that can be established early and preserved across stages. In this work, the term semantic does not refer to category level recognition or ontology level reasoning. Instead, it refers to directional semantics: how users infer, from object structure and view dependent cues, which direction is most interpretable and actionable for manipulation[23, 34, 39].

In SnapAxis, this directional grounding can come from two broad sources: intrinsic structural cues of the object and view dependent visible cues. Intrinsic cues arise from relatively stable directional organization that users can naturally attribute to the object, such as a canonical object centered direction or a strong elongation trend. View dependent cues arise from what is currently visible in the captured image, such as the dominant contour trend or the most prominent visible face. Based on our observations of how users inferred object direction during early manipulation, together with prior literature on object centered direction[7, 20], elongation based axis[43], contour based orientation cues[19], and visible face references[56], we summarize four common grounding cases: an intrinsic object level direction, which reflects the object's overall directional organization; a salient elongation direction, which follows a visually dominant long axis structure; a view induced contour direction, which reflects the main directional trend of the visible outline; and a visible face normal direction, which is derived from the dominant visible surface when no clear long axis cue is available. We describe these as common grounding cases for the SAH rather than as a standalone semantic taxonomy.

Accordingly, SnapAxis does not implement a separate semantic classifier for these cases. Instead, it establishes the SAH through a lightweight heuristic based on the object's screen space exposure and the presence of visually reliable directional evidence. Let denote by a_i the normalized screen space area of the object, computed as the ratio between its 2D bounding box area and the full image area. The exposure thresholds and fallback rules used here were determined empirically during formative prototyping and

pilot trials, with the goal of balancing directional stability against the limited visible evidence available under distance and occlusion. When $a_i > 0.35$, the object is considered sufficiently exposed to support a stable directional reference. In this case, the system prioritizes stronger structural cues, such as a salient elongation direction or a stable intrinsic object level direction. When $0.18 \leq a_i \leq 0.35$, exposure is only partial, so the system relies on whichever local directional cue remains visually interpretable, typically a visible face normal direction or a still discernible elongation trend. when $a_i < 0.18$, the visible evidence is too limited to support a stable structural interpretation, and the SAH falls back to a view-induced contour direction in the current view.

In this way, the SAH is always grounded in the strongest directional cue that remains visually interpretable under the current viewing condition, rather than in an arbitrary control axis. This design keeps the SAH computationally lightweight while preserving its role as a stable manipulation reference across stages. Rather than requiring users to reinterpret new control mappings at each stage, the SAH provides a directionally meaningful handle that can be previewed during acquisition, externalized during coarse manipulation, and preserved during fine manipulation.

3.3 SnapAxis: An Axis based Interaction Framework

As shown in Figure 2, SnapAxis is an axis based interaction framework for smartphone based MR object manipulation. Its central design idea is to organize acquisition, placement, and refinement around a shared object centered reference, the SAH. It does not treat these stages as separate interaction problems with unrelated control mappings. Instead, the framework emphasizes maintaining a stable and interpretable reference as interaction progresses. From this perspective, the challenge is not only to reach a distant or occluded object, but also to preserve a usable reference throughout the workflow.

To operationalize this idea, SnapAxis consists of three connected stages: ZoomCapture, Coarse Manipulation, and Fine Manipulation. In ZoomCapture stage, the user uses smartphone camera to capture the current view of the target with the smartphone and enlarges the object's visible exposed region through zooming. This process not only supports target search, localization, and confirmation, but also provides a view consistent reference for subsequent interaction, thereby establishing the visual basis for SAH formation. After confirming, the framework then enters the Coarse Manipulation stage, in which the user specifies the location of the placement of the object, sketches its principal axis on the screen with the reference axis, and adjusts its depth to perform initial translation and rotation. Through this process, the SAH is gradually externalized, and the SAH is established as an object centered manipulation reference. After confirming the axis and generating the axis endpoints, the user enters the fine manipulation stage. In this stage, the user refines the final transformation of the object through a series of operations performed on the sketched axis. Specifically, different manipulations of the axis endpoints enable fine grained translation, rotation, and scaling of the object. Users may return from Fine Manipulation to Coarse Manipulation when a new axis needs

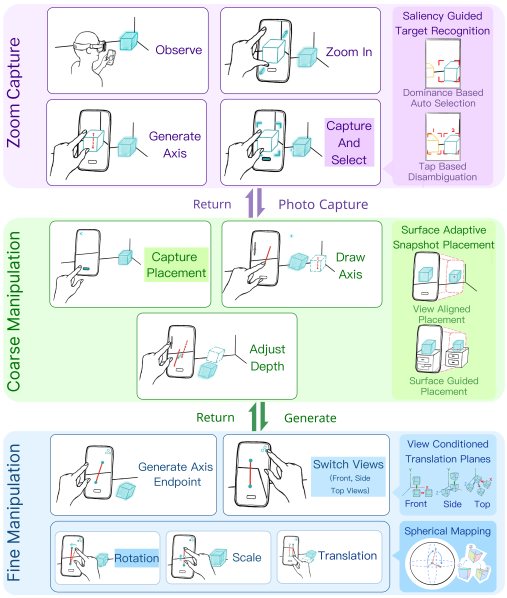


Figure 2: The SnapAxis framework. The SAH is established in Zoom Capture through saliency guided target recognition, then reused in Coarse Manipulation, where a stroke sets the axis and a depth slider adjusts distance, and in Fine Manipulation, where dragging its endpoints refines rotation, scale, and translation. Return arrows indicate reversible stage transitions.

to be specified, allowing the framework to support both iterative correction and progressive refinement.

Together, these stages structure object manipulation as a process of establishing, externalizing, and preserving a shared object centered reference, rather than repeatedly redefining control mappings across independent interaction modes. Throughout the workflow, the HMD serves as the primary MR viewport for observation and verification, while the phone acts as the input surface and secondary display, supporting zoom capture, stroke drawing, depth adjustment, view switching, and endpoint dragging. Transitions between the three stages are triggered by on screen buttons on the phone, and users can move both forward and backward between adjacent stages at any time through the Back button. The following sections describe each stage in detail.

3.4 ZoomCapture: Target Acquisition

Building on the design guideline 1, ZoomCapture turns the smartphone into a target acquisition interface. It supports both distant search and close inspection by allowing users to zoom before capture. By enlarging the visible exposed region of the target before commitment, it helps users better evaluate what is being selected. This reduces reliance on ambiguous depth cues during cross scale targeting. Figure 2 top illustrates the ZoomCapture stage.

The user first observes the MR scene through the smartphone view and then moves the device to locate the target. When the target is identified but still appears visually small, the user can zoom in on the scene so that more of the target becomes visible

on the screen, making the subsequent capture more reliable. Once the user is satisfied with the current view, the zoomed image is captured to initiate target selection. After the target is confirmed, the system displays an initial preview of the SAH in the captured view, represented as the object's principal axis under the current viewpoint. This view dependent preview serves as the first visible form of the handle, providing the user with an interpretable initial directional reference for subsequent interaction.

To support target identification in the captured zoomed view, SnapAxis performs a lightweight frame analysis based on the 2D bounding boxes of visible objects. These 2D boxes are computed by projecting each virtual object's 3D mesh bounding box onto the current phone camera view. For each object, the system computes its normalized screen space area a_i (i.e., the 2D bounding box area divided by the image area). When the top ranked candidate is sufficiently dominant, the system applies Dominance Based Auto Selection using two tunable criteria: (1) an absolute dominance criterion, $a_1 \geq 0.18$, and (2) a top 2 separability criterion, $a_1/a_2 \geq 1.35$. These thresholds were determined empirically through pilot studies to balance automatic selection reliability against false positives. If either criterion is not satisfied, the system falls back to Tap Based Disambiguation. In this case, up to two candidate objects are overlaid in the ZoomCapture view with numeric labels, allowing the user to confirm the intended target with a single tap, thereby minimizing additional interaction overhead.

3.5 Coarse Manipulation: Placement and Axis Specification

Building on Design Guidelines 2 and 3, Coarse Manipulation externalizes directional intent through axis sketching. It organizes placement, orientation specification, and depth adjustment into an initial manipulation stage. The goal of this stage is not final precision, but rapid spatial structuring. Before entering Fine Manipulation, users first establish an approximate object position and directional reference. Figure 2 middle illustrates the Coarse Manipulation stage.

The axis is drawn because a stroke is the simplest way for the user to express both the object's intended position and direction, which suits the early manipulation stage where the overall pose is still being established. Two alternative axis choices were also considered and set aside. World axes are independent of any single object, and aligning to them takes rotation and scaling outside the object relative frame identified in Design Guideline 2. Fixed bounding-box axes are purely geometric, and for non axisymmetric objects the functional axis often coincides with none of them.

The stage begins with capture placement, where the user captures the desired placement region and the system initializes the target object from the current snapshot. Building on the handle preview introduced in ZoomCapture, the user then binds and externalizes this previewed handle by drawing a stroke on the screen. The stroke position determines coarse placement, while the stroke direction confirms and instantiates the handle's directional intent, which is mapped to a target 3D orientation through spherical mapping. The system rotates the object accordingly and updates the in scene preview in real time. In this way, the stroke does not define an entirely new axis from scratch, but anchors the previewed handle to the current placement and turns it into an explicit object centered manipulation reference. The user can then refine the coarse result with a depth slider. Under distance and occlusion, directly moving an object in depth often requires simultaneous judgment of depth, lateral position, and their relation to the target. By contrast, the depth slider reframes depth as a near far adjustment relative to the current view, making coarse placement more stable and interpretable. Axis specification and depth adjustment can be alternated until the user is satisfied.

To support object placement, the user can zoom in and out to refine the framing region and capture a snapshot that specifies the desired placement location of the object. When a horizontal surface is detected through AR plane detection on the phone, our method estimates the initial placement by intersecting the smartphone camera's center viewing ray with the detected plane. To prevent interpenetration with real world geometry, bounding box collision handling is enabled so that the virtual object is not placed inside physical objects. If no horizontal surface is detected, the object is instantiated on a virtual projection plane perpendicular to the camera's center viewing direction. This snapshot based placement establishes a viewpoint consistent initial position and provides a reference plane for subsequent manipulation. When a horizontal surface is available, all stroke based operations constrain the object to remain on that surface. Otherwise, all stroke based adjustments are performed within the virtual projection plane.

3.6 Fine Manipulation: Axis Oriented Refinement

Building on Design Guidelines 2 and 3, Fine Manipulation performs precise refinement around the already established SAH. Rather than redefining the control reference, this stage operates on the same handle to support low coupling adjustment of translation, rotation, and scaling. In this way, it preserves coarse to fine stage while allowing users to progressively refine the result obtained from Coarse Manipulation. Figure 2 bottom illustrates this stage.

After the coarse result is confirmed, the sketched stroke is converted into a manipulable principal axis with two endpoints. Users can then switch among front, side, and top views. Each view switch automatically repositions and reorients the camera around the object center, providing a stable visual reference for subsequent operations. The lower endpoint controls translation, whereas the upper endpoint controls rotation and scaling. Translation is always constrained to the 2D plane aligned with the current view, which helps reduce depth ambiguity during refinement. Scaling is performed by dragging along the direction of the principal axis, whereas rotation is achieved through a spherical mapping mechanism. Users can switch freely among translation, rotation, and scaling modes, and may return to the view switching interface at any time.

Translation is always performed within a 2D plane aligned with the current view, rather than in unconstrained 3D space. Specifically, the current view determines the active manipulation plane. In the front view, dragging the lower endpoint translates the object in the x-y plane; in the side view, it translates the object in the z-y plane; and in the top view, it translates the object in the x-z plane. This view based aligned design also helps resolve depth ambiguity during refinement.

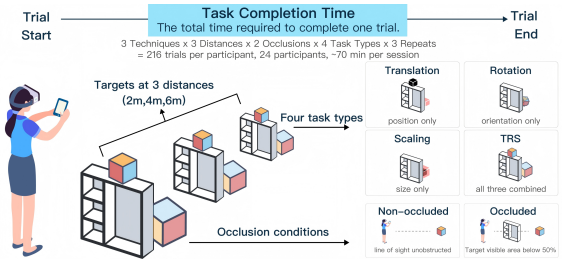


Figure 3: Experimental design showing the target layout and interaction phases. Users perform targeting and docking. The trial repeats across distances and techniques.

Rotation is implemented through spherical mapping, which projects the user's drag direction onto the tangent plane of a sphere and incrementally updates the object's orientation around the principal axis. To further improve stability, SnapAxis adopts an incremental update strategy with a fixed gain factor of $g_r = 0.6$ radians per unit normalized drag distance. This value was determined through pilot testing. The reference frame is automatically reset once the user confirms the operation, which helps maintain smooth and predictable control while reducing the risk of overshooting.

4 User Study

To evaluate the performance of SnapAxis for MR object manipulation under distance and occlusion, we conducted a controlled within subject smartphone based user study(figure 3).

4.1 Participants and Apparatus

We recruited 24 participants (12 female, 12 male; age 21–32, $M = 25.4$, $SD = 3.1$). All participants were right handed, had normal or corrected to normal vision, and reported no motor impairments. All were familiar with smartphones, and 18 had prior VR/MR experience, although none had previously used the techniques evaluated in this study. Participants were recruited from a local university and the surrounding community and received monetary compensation. The study was approved by the institutional ethics committee, and informed consent was obtained from all participants. The system was implemented in Unity 2022.3.26f2c1 and deployed on a Pico 4 Ultra headset with external trackers. We logged object transformations, user input events, and task state transitions throughout the study. All conditions were implemented within the same Unity project and shared a common codebase, differing only in the mapping from input to manipulation.

4.2 Experimental Design

Task Participants performed a standard 6DoF docking task in a $5 \times 8 \times 3$ m physical space, in which they manipulated a 3D object to match a given target pose [2]. The task included four types: Translation, in which only the object position was adjusted; Rotation, in which only the orientation was adjusted; Scaling, in which only the object size was adjusted; and TRS, in which position, orientation, and scale could all be adjusted.

In addition to task type, we introduced Distance and Occlusion as independent variables. Distance refers to the spatial distance between the participant and the manipulable object. We included three distance levels: 2 m, 4 m, and 6 m,to represent near, mid, and far field interaction ranges. Occlusion was manipulated as a binary factor (non occluded vs. occluded) to examine how performance changed when visual obstruction was introduced [11]. Non occluded means the target is unobstructed. Occluded means one opaque box between viewpoint and target, cutting visible area to below 50%. This design allowed us to compare SnapAxis against both a traditional far field baseline and a representative smartphone based baseline, while using the non occluded condition as a reference for quantifying the additional performance cost introduced by occlusion.

To ensure a fair comparison, we implemented a lightweight state machine to manage repeated trials across task types. At the beginning of each trial, the state machine updated the target object pose according to the assigned task type while keeping the distance and occlusion settings fixed. During the task, the manipulable object was highlighted in blue, and a semi transparent ghost object was displayed at the target position and pose to guide alignment. The initial position of the manipulable object remained fixed across repeated trials of the same task type.

Conditions We employed a within subject design with three techniques. We compared SnapAxis (EC) against two baselines: a conventional ray based interaction method (CC1)[13] and an MR adapted HandyCast method (CC2) [26]. The ray based method served as a standard far field interaction baseline, while HandyCast served as a smartphone based manipulation baseline. CC1 reproduced PRISM's original velocity-dependent ray mapping, with angular velocity computed over a 200-ms interval and a scaling constant of 20°/s. CC2 reproduced HandyCast's original nonlinear pose transfer mapping ($\lambda = 2.5$) and relative gain accelerated touch mapping. Because HandyCast does not natively support scaling, we added a two fingers touch gesture to enable scaling and ensure a fair comparison. This gesture is common on smartphones and consistent with HandyCast's interaction style. All conditions were implemented in the same MR system and evaluated under identical task, distance, and occlusion settings.

Procedure Participants first completed a five minutes training session to familiarize themselves with the interaction mappings between the smartphone and the headset. Trials were grouped into sessions by task type, with a two minute break between conditions to reduce fatigue. For each participant, the order of conditions was randomized with equal probability. After each condition, participants completed a subjective questionnaire. Each participant completed 3 Conditions × 3 Distances × 2 Occlusions × 4 Tasks × 3 Repeats = 216 trials. Each trial lasted approximately 15 s on average, and the full study lasted about 70 minutes per participant.

Each trial began with target observation and ended when the participant pressed *Confirm*, which triggered success checking. In the single factor tasks, only the task relevant degree(s) of freedom were enabled, while the remaining dimensions were initialized to the target value and kept fixed throughout the trial. Accordingly, success was evaluated only on the manipulated dimension(s). For Translation trials, success required the positional offset to be below 20% of the object's visual angle. For Rotation trials, success required the angular error to be within 15°. For Scaling trials, success required the relative scale error to be within 10% of the target scale,

i.e., $|s - s_{\text{target}}|/s_{\text{target}} \leq 0.10$. For TRS trials, all three criteria had to be met simultaneously, after which the system advanced automatically. Thresholds were derived from prior MR docking criteria[51] and refined through pilot testing. We tested five levels for each transformation and selected 20% translation, 15° rotation, and 10% scaling to balance technique discrimination and completion rate.

4.3 Metrics

We adopted a comprehensive set of objective and subjective measures to evaluate participant performance. These measures were organized into three complementary dimensions: Task Accuracy, Task Efficiency, and User Experience. Together, they capture not only whether a technique enables successful manipulation, but also how accurately and efficiently users complete the task, as well as how they subjectively perceive the interaction process.

Task Accuracy.

- **Task Relevant Alignment Error** (↓): The final translation error, rotation error, scaling error, or combined TRS error.

Task Efficiency.

- **Task Completion Time** (↓): The total time required to complete one trial.
- **Target Acquisition Time** (↓):The time from trial onset to successful target acquisition.
- **Phone Path** (↓): The cumulative 3D travel distance of the smartphone during a trial, reported in meters.
- **Head Pitch/Yaw** (↓):The cumulative absolute angular change of the headset around the vertical axis and lateral axis.

User Experience.

- **NASA-TLX** (↓): Participants' perceived workload during task completion.
- **SUS** (↑): Participants' overall usability rating of the technique.
- **Borg CR10** (↓): Participants' perceived physical exertion during the task.
- **Qualitative feedback**: Post study interview responses concerning ease of use, perceived efficiency, perceived accuracy, and control confidence.

4.4 Results

We analyzed the objective measures using repeated measures ANOVA, with Technique, Distance, and Occlusion treated as within subject factors. When the sphericity assumption was violated, Greenhouse Geisser correction was applied. Significant effects were followed by Bonferroni corrected post hoc pairwise comparisons. Effect sizes are reported as partial eta squared (η_p^2), with .01, .06, and .14 interpreted as small, medium, and large effects, respectively. For subjective workload, we analyzed the overall NASA-TLX score[21] and SUS scores[6] using one way repeated measures ANOVA with Technique as the within subject factor, while Borg CR10 ratings[3], being ordinal, were analyzed with Friedman tests followed by Bonferroni corrected Wilcoxon signed rank tests. Open ended interview responses were analyzed qualitatively through descriptive summarization of recurring themes and representative participant feedback. In the tables, significance levels are denoted as $^{*}p < .05$, $^{**}p < .01$, and $^{***}p < .001$. The TRS task most closely reflects real world 3D object manipulation, we treat it as the primary focus of analysis in the main text. The single factor tasks (T, R, and S) are mainly used to explain the source of the observed differences across conditions.

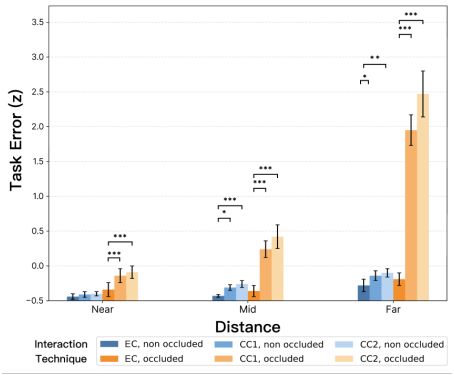


Figure 4: Mean task relevant alignment error across distances, occlusion levels, and techniques.

4.4.1 ***Task Accuracy.*** As shown in Figure 4, EC significantly reduced task error relative to both CC1 and CC2 under most mid and far distance conditions(main effect of Technique, $F(2, 46) = 32.14, p < .001, \eta_p^2 = .58$). At the mid distance, EC outperformed CC1 under both non occluded and occluded conditions ($p < .05$ and $p < .001$), and also outperformed CC2 under both conditions ($p < .001$ for both). The same pattern was observed at the far distance, where EC again significantly outperformed CC1 ($p < .05$ and $p < .01$) and CC2 ($p < .001$ for both). This suggests that EC's accuracy advantage became more evident as interaction difficulty increased. By contrast, at the near distance, EC showed lower task error but a weaker advantage: under the non occluded condition, differences from CC1 ($p = .214$) and CC2 ($p = .118$) were not significant, whereas under the occluded condition EC significantly outperformed both baselines ($p < .001$ for both). Under the near non occluded condition, where alignment was easier overall, differences between techniques were limited.

By comparison, CC1 and CC2 did not differ significantly in task error in any condition (TRS2: $p = .731/.664$; TRS4: $p = .284/.173$; TRS6: $p = .352/.091$), indicating comparable overall TRS alignment performance between the two baselines. Across all techniques, task error was significantly higher under occlusion than non occlusion (main effect of Occlusion, $F(1, 23) = 156.83$,p<.001, $\eta_p^2 = .87$), indicating that occlusion introduced additional difficulty for alignment. Task error also increased with distance (main effect of Distance, $F(2, 46) = 48.72, p < .01, \eta_p^2 = .68$), indicating that larger distance generally increased task difficulty.

To further identify the source of this pattern, we analyzed the T, R, and S subtasks. The largest main effect of Technique was observed for Translation ($F(2, 46) = 31.54, p < .001, \eta_p^2 = .58$), whereas Rotation and Scaling showed smaller effects ($\eta_p^2 = .017$ and $\eta_p^2 = .019$, respectively). This indicates that Translation contributed most strongly to the observed TRS accuracy differences.

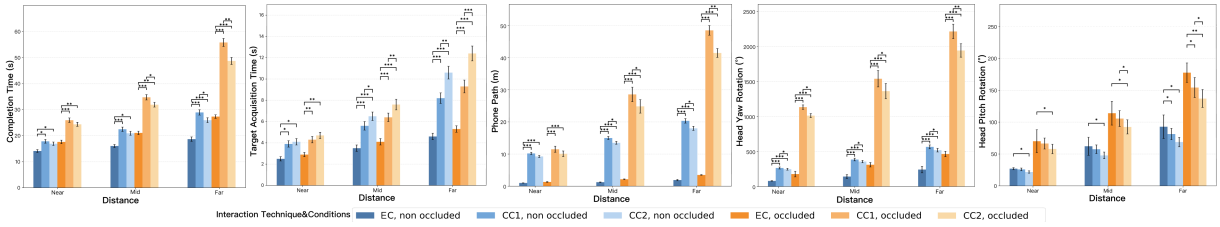


Figure 5: Mean task efficiency across interaction techniques, distances, and occlusion conditions. From left to right, the panels report completion time (s), target acquisition time (s), phone path (m), head yaw rotation (°), and head pitch rotation (°).

4.4.2 ***Task Efficiency.*** As shown in Figure 5, EC consistently demonstrated higher efficiency than CC1 and CC2 across distance and occlusion conditions. For completion time, there was a significant main effect of Technique ($F(2, 46) = 28.46, p < .001, \eta_p^2 = .55$), as well as significant Technique × Distance ($F(4, 92) = 9.84, p < .001, \eta_p^2 = .30$) and Technique × Occlusion ($F(2, 46) = 15.72, p < .001, \eta_p^2 = .41$) interactions. Phone path and head yaw showed similar patterns, indicating that EC not only enabled faster task completion, but also required less device movement and lower horizontal visual search effort. By contrast, for head pitch, EC was slightly higher than CC1 and CC2 at mid and far distances, but the main effect of Technique was not significant ($F(2, 46) = 2.91, p = .065, \eta_p^2 = .11$). Overall, the advantage of EC was reflected primarily in completion time, phone movement, and horizontal head rotation.

CC2 was more efficient than CC1, but the difference was smaller. For completion time, the CC1–CC2 difference progressed from $p = .362$ / $p = .184$ under the near condition to $p = .091$ / $p < .05$ under the mid condition, and further to $p < .05$ / $p < .01$ under the far condition. This suggests that the difference between the two baselines was limited at near distance, but became more apparent at mid and far distances. Similarly, CC2 showed lower phone path and head yaw than CC1, suggesting greater efficiency.

Increasing distance raised completion time, phone path, and head yaw across all techniques. For example, completion time showed a significant main effect of Distance ($F(2, 46) = 112.38, p < .001, \eta_p^2 = .83$). This indicates that larger interaction distances generally increased the procedural cost of task completion. Similarly, introducing occlusion increased completion time, phone path, and head yaw. For completion time, the main effect of Occlusion was significant ($F(1, 23) = 136.27, p < .001, \eta_p^2 = .86$). This indicates that occlusion imposed additional efficiency costs across techniques.

Further analysis showed a significant main effect of Technique on Target Acquisition Time ($F(2, 46) = 31.27, p < .001, \eta_p^2 = .58$). EC was significantly lower than two baselines, indicating that its efficiency advantage was already present during target acquisition and confirmation. This pattern was consistent with the single factor task results, in which Translation showed the strongest efficiency related advantage. Overall, efficiency differences emerged not only during later manipulation but also during early target acquisition.

4.4.3 ***Robustness Across Distance Under Occlusion.*** We analyzed TRS performance across distances under occlusion. From 2 to 6 m, completion time increased from 25.96 to 55.84 s for CC1 and from 24.38 to 48.73 s for CC2, but only from 17.62 to 27.35 s for EC. At 6 m, EC was 51% faster than CC1 and 44% faster than CC2.

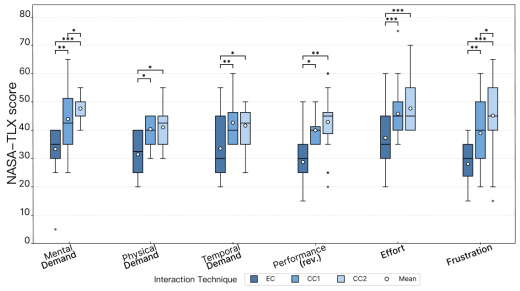


Figure 6: NASA-TLX Ratings for each interaction technique on six subscales.

The phone-path difference was even larger: EC required 3.49 m of device motion, compared with 48.52 m for CC1 and 41.47 m for CC2 ($p < .001$). Thus, under occlusion, both baselines degraded with distance, while EC remained stable, widening its advantage.

4.4.4 ***User Experience.*** As shown in Figure 6, ANOVA on the overall NASA-TLX score revealed a significant main effect of Technique ($F(2, 46) = 26.18, p < .001, \eta_p^2 = .53$). Bonferroni corrected post hoc comparisons showed that EC yielded significantly lower workload than both CC1 and CC2 (both $p < .001$), while the difference between the two baselines was not significant ($p = .27$).

Descriptively, the clearest subscale differences appeared in Effort, Frustration, and Mental Demand, where EC consistently received lower ratings than both baselines. Qualitative feedback showed a similar pattern: (16/24) participants described EC as easier to use and less effortful, especially under distant and occluded conditions. Some participants noted that EC helped them understand target orientation more quickly and reduced unnecessary searching. As one participant put it, "after capturing it, the target felt fixed inside the phone, so I stopped searching for it in space," reflecting how the smartphone shifted from an auxiliary input device to a view-grounded reference. A smaller subset (5/24) preferred the baselines in near-distance conditions because of their familiarity, and (3/24) mentioned a slight learning curve for EC. SUS and Borg CR10 showed the same ordering. SnapAxis achieved higher usability than both baselines (SUS 82.3 vs. 63.1 and 60.4; $F(2, 46) = 19.7, p < .001, \eta_p^2 = .46$, both Bonferroni corrected $p < .01$) and lower perceived exertion (Borg median 2.0 vs. 5.0 and 4.5; $\chi^2(2) = 21.6, p < .001$, both $p < .001$), with no significant difference between the two baselines. Overall, these results suggest

that EC reduced perceived workload and improved user experience, particularly in more challenging conditions.

5 Discussion

The user study results strongly support the effectiveness of EC. In terms of task accuracy, the results suggest that the core challenge of MR 3D manipulation is not merely reaching the target pose, but sustaining reliable depth judgment and stable multi DoF refinement under distance and occlusion. Although all techniques supported basic 3D alignment, they differed substantially in how depth reference and multi DoF control were established. CC1 relied on direct manipulation and continuous visual correction. This was workable under near and non occluded conditions, where depth judgments were relatively stable. However, the technique became less reliable as distance increased. Under those conditions, users had to correct translation, rotation, and scale while working from weaker depth evidence. As a result, adjustment along one dimension often disturbed the others, leading to overshoot and repeated readjustment. CC2 partially alleviated depth issues through pose and touch based virtual hand control, which amplified small smartphone movements into remote manipulation. However, it still required users to continuously coordinate position, orientation, and scale, and thus remained vulnerable to limited visibility and depth ambiguity under distance and occlusion. By contrast, EC reduced control coupling by establishing an object centered handle before fine refinement. ZoomCapture and coarse manipulation first helped users form a stable placement and directional reference. Fine manipulation then operated around the same SAH rather than switching to a new control logic. The sketching step was lightweight (median stroke 1.5 s; single stroke in 96% of trials). This made refinement more structured and interpretable, which in turn improved alignment accuracy. At the same time, EC's advantage was weaker under the near non occluded condition, where differences from the baselines were not significant. This suggests that when the target is close and fully visible, the scene already provides sufficiently stable perceptual support for accurate alignment, allowing even the baseline techniques to perform well, and the benefit of a stable object centered handle is therefore less amplified than under mid /far distance or occluded conditions. In terms of task efficiency, the observed differences mainly reflect the costs of target acquisition and viewpoint adjustment. Whereas accuracy reflects whether users can reliably achieve alignment, efficiency reflects the procedural cost required to do so. For CC1, interaction relied on continuous adjustment, so under far field conditions users required longer target confirmation and more trial and error operations, leading to longer completion times and phone paths as they moved the device over a wider range to find an effective operating position. Under occlusion, additional head movement and viewpoint repositioning were needed to restore visibility, further prolonging the task. CC2 improved remote manipulation through virtual hand control, but under distance and occlusion users still had to repeatedly adjust device position and viewing angle before reaching an effective interaction state. EC's efficiency advantage emerged early in the interaction process. During ZoomCapture, users could enlarge the target's visible region before selection, which made search, localization, and confirmation faster. During coarse manipulation, the technique established the handle that was later reused for refinement. As a result, fine manipulation proceeded with less remapping and less trial-and-error. Together, these benefits reduced completion time, target acquisition time, phone movement, and horizontal visual search effort. One exception was head pitch, for which EC did not show the same advantage. This pattern may be related to users shifting attention between distant targets and the nearby handheld device. While EC reduced completion time, phone path, and head yaw, users still needed to look downward to inspect smartphone based feedback during acquisition and coarse manipulation. As a result, the efficiency advantage of EC was less evident in head pitch than in the other efficiency measures. These objective advantages further translated into a better user experience. By improving both target acquisition and later refinement, EC reduced the physical and cognitive effort required to complete the task, which was also reflected in its lower NASA-TLX scores. Taken together, these results suggest that the *SnapAxis* framework, structured around the SAH, provides clear advantages for MR 3D object manipulation under distance and occlusion.

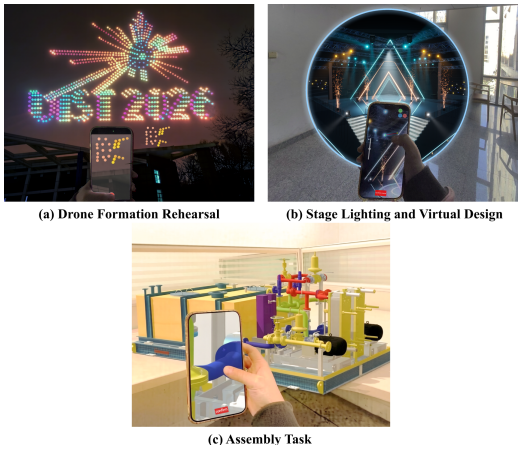


Figure 7: Application scenarios. (a) Distant target acquisition in drone formation rehearsal. (b) Midair directional alignment in stage lighting and virtual design. (c) Structure aware alignment in assembly tasks under occlusion.

6 Application Scenarios

This section illustrates representative application scenarios for SnapAxis. We focus on three task situations in which smartphone assisted MR manipulation becomes especially challenging: selecting distant targets, aligning suspended objects in midair, and refining partially occluded objects in cluttered 3D environments.

6.1 Distant Target Acquisition

Reliably selecting small or distant targets is a prerequisite for any subsequent alignment operation. However, when the target occupies only a few pixels in the view and depth cues are weak, target selection becomes unstable. Consider the application of drone formation rehearsal: an operator may need to select a specific drone from a long distance (Figure 7 (a)). In such scenarios, traditional ray casting or gaze based selection methods are prone to drift, often

leaving users in a state of uncertainty they may select something, but cannot confidently determine whether it is the intended target. To address this issue, our method introduces ZoomCapture, which enlarges the region of the target to support more precise selection. By improving target discriminability, this helps reduce downstream alignment failures caused by unstable target acquisition.

6.2 Midair Directional Alignment

When an object is suspended in space and the ground plane cannot provide an effective reference, specifying its intended orientation becomes particularly difficult. A representative example is stage lighting and virtual design (Figure 7 (b)), where lighting fixtures often need to be positioned and oriented in midair. In such cases, ground-based dragging provides little support for expressing directional intent and often leads to repeated trial-and-error adjustments. By representing orientation through an object-centered principal axis, our method provides users with a directly manipulable directional reference. This reference remains stable even under distance or limited visibility, supporting more consistent alignment when the scene lacks sufficient environmental cues.

6.3 Structure Aware Alignment Under Occlusion

A third task class involves alignment under partial occlusion, where users must manipulate an object based on incomplete visual evidence. In these tasks, failure often stems not from the absence of fine manipulation, but from the lack of a stable structural reference before refinement begins. For example, in assembly or setup scenarios, surrounding structures and neighboring components often occlude the target object, making it hard to accurately infer the object's true boundaries and orientation from a single viewpoint. Consider the application of assembly task (Figure 7 (c)) .As a result, users may align the object based only on its visible fragments, misjudging its actual extent or spatial direction. SnapAxis addresses this challenge by allowing users to ground manipulation in an object centered axis before precise adjustment. Rather than requiring refinement directly on incomplete visual input, the workflow helps users first recover a usable structural reference and then perform alignment relative to that reference. This makes subsequent manipulation more stable and interpretable, especially in visually cluttered environments.

7 Limitations and Future Work

Our method has four main limitations. First, our comparison covers two baselines selected to represent the dominant far field and smartphone based control paradigms. Broader comparisons with other smartphone based and proxy based techniques remain future work. Second, although SnapAxis reduces interaction ambiguity by progressively establishing SAH, part of the current method still depends on scene related configuration. In particular, when no suitable environmental plane is available, the position of the control plane remains a task sensitive parameter and is currently determined through empirical tuning. Future work could explore more adaptive scene analysis strategies that automatically estimate suitable control plane placement based on object distance, spatial layout, and viewing conditions. Third, like other headset smartphone collaborative interaction techniques, our method may still introduce additional perceptual burden associated with video see through usage. Users must repeatedly shift visual focus between distant 3D targets and the nearby handheld smartphone screen. This cross depth, cross device switching may increase perceptual conflict, cognitive load, and visual fatigue, and the phone screen itself is viewed through passthrough rather than directly, so its reduced effective resolution and clarity may have made small elements such as the axis endpoints and the depth slider harder to resolve, a factor that applies equally to both phone based conditions and should be considered when interpreting absolute task times. Although the progressive workflow reduces how often users must read the phone, this reading load is reduced rather than eliminated. Future work could reduce dependence on the smartphone by moving more feedback into the MR view, adopting higher contrast phone UI and audio cues, or gradually transitioning toward fully in headset interaction. Finally, the method assumes relatively static scenes and may be less effective for moving, deforming, or dynamically occluded objects. Future work could incorporate continuous tracking, dynamic principal axis reestimation, and predictive control.

8 Conclusion

We presented SnapAxis, an axis based interaction framework for smartphone based MR object manipulation under distance and occlusion. Rather than treating target acquisition, coarse placement, and fine refinement as separate interaction problems, SnapAxis organizes them around a shared the SAH. Through ZoomCapture, coarse manipulation, and axis oriented refinement, the framework progressively establishes, externalizes, and preserves this handle across stages. Our user study showed that SnapAxis improved task accuracy, task efficiency, and perceived workload relative to two baselines. In particular, its advantages became more pronounced under mid and far distance and occluded conditions, where maintaining stable references and reducing control coupling are especially important. More broadly, our findings suggest that effective MR manipulation depends not only on reaching the final pose, but on how interaction supports users in forming and maintaining a stable object centered reference throughout the workflow.

Acknowledgments

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